

Tang Zhice

Chongqing, China

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EDUCATION

Chongqing University, Chongqing, China

Sep. 2023 - Jun. 2027 (Expected)

Bachelor of Artificial Intelligence, School of Big Data & Software Engineering

GPA: 3.45/4.0, **Rank:** 7/53

Relevant Coursework: Deep Learning (95), Statistical Analysis (96), Data Structures (93), Programming Techniques (96)

RESEARCH INTERESTS

Computer Vision Multimodal Large Language Models Image and Video Generation

Embodied AI Medical Image Analysis

RESEARCH EXPERIENCE

Project: *"GASP-SAM: Lightweight Geometry-Aware Semantic Prompting for Abdominal Organ Segmentation"*
Dec. 2024 - May. 2025

Project Leader

- Led a research project on adapting SAM ViT-B for box-prompted abdominal multi-organ CT segmentation.
- Proposed GASP-SAM, a parameter-efficient two-stage framework combining LoRA-based encoder adaptation, organ-aware semantic prompting, geometry-guided residual fusion, and local residual refinement.
- Designed a BioMedCLIP-based semantic prompt learner and TextCIM module to inject anatomical organ priors into SAM's class-agnostic visual features.
- Developed a geometry guidance module that predicts boundary, center, and signed-distance cues and uses them as inference-time structural priors for decoder feature modulation.
- Built a compact local residual refiner that corrects boundary errors only within ROI regions, avoiding unnecessary modification of reliable coarse predictions.
- Achieved 90.86% mean Dice on the FLARE22 common eight-organ subset and 87.82% mean Dice on Synapse with only 8.10M trainable parameters, including 0.72M in the local refiner.
- Implemented the full training and evaluation pipeline in PyTorch, including LoRA adaptation, semantic-geometry supervision, staged training, and 3D reconstruction-based evaluation.

Project: *"LiftCut Tracker: AI-Powered Fitness and Fat-Loss Tracking Platform"*

Mar. 2026 - Now

Project Developer

- Built and deployed a clean, ad-free fitness tracking web application that combines training plans, workout logs, nutrition records, body metrics, and AI-generated coaching workflows. The project was designed as a real product rather than a static demo, with authenticated user data, server-side AI APIs, persistent database storage, and a guest-to-account migration path.
- Implemented a Next.js App Router architecture with TypeScript, Tailwind CSS, shadcn/ui, Zustand, Zod validation, next-intl localization, Recharts dashboards, and PDF export through jsPDF.
- Deployed server-side AI API routes for training-plan generation, nutrition-plan generation, plan saving, and AI history management, keeping the DeepSeek API key isolated in server-only environment variables.
- Provisioned a Supabase backend with Auth, Postgres, Storage, row-level security policies, relational training/nutrition schemas, indexed user timelines, and an avatars bucket for profile media.

- Designed a structured AI workflow: user profile and constraints are validated, DeepSeek returns schema-checked JSON, users preview and edit the result, and only confirmed plans are persisted.
- Built guest mode with localStorage persistence, daily AI quota control, and one-click migration of training plans, workout logs, food logs, body metrics, and AI history into a formal account.

Project: “Real-time Face Recognition System”

Aug. 2024 - Sep. 2024

Project Developer

- Designed and implemented an end-to-end face recognition system using Python, TensorFlow/Keras, OpenCV, and Flask, covering data collection, model training, and web deployment.
- Built a real-time face data collection tool with OpenCV and Haar Cascade classifiers, enabling automated face detection, cropping, and per-person dataset organization.
- Constructed a two-layer CNN for multi-class face classification with max-pooling, a 1024-unit fully connected layer, Adam optimizer, and categorical cross-entropy loss.
- Deployed the trained model as a Flask-based web application supporting both image-upload recognition and real-time video-stream recognition.
- Implemented preprocessing and system modules including grayscale normalization, adaptive interpolation, train-test split, exception handling, and resource management.

INDUSTRY EXPERIENCE

CITIC Securities, Beijing, China

Jan. 2024 - Feb. 2024

Intern(Remote) /Full-time

- Assisted in industry research by collecting, cleaning, and organizing market and company data from multiple sources.
- Conducted data analysis and visualization to support sector trend tracking and investment research.
- Contributed to research reports by summarizing industry developments, comparing company performance, and presenting key findings in a structured and concise manner.

ACTIVITIES & AWARDS

The Second Prize Scholarship, Chongqing University

Apr. 2024

The Third Prize Scholarship, Chongqing University

Apr. 2025

Merit Student of Chongqing University

Apr. 2025

Captain, National Flag Guard Team, Chongqing University

Sep. 2023 - Present

Led routine training, team coordination, and ceremonial activities, demonstrating leadership, discipline, and teamwork.

Supported major campus events requiring formal presentation and institutional representation.

LANGUAGES & SKILLS

Languages: Mandarin (Native), English (IELTS: 6.5)

Programming: Python, C++

Frameworks/Tools: PyTorch, TensorFlow/Keras, OpenCV, Flask, GitHub, Hugging Face

Other Skills: Data Analysis, Data Visualization, Microsoft Office (Word, Excel, PowerPoint)